

UCCC2063 ALGORITHMS ANALYSIS**SECTION A (Answer All Questions)**

Q1. [Asymptotic Notations]

(a) Express the following functions in terms of Ω notation.

(i) $n^{2023} \log n^{2024} + n^{2025}$ (1 mark)

(ii) $(7n)^{98} + 40n(\log_2 n)^{123} + 0.01(2^n) + 2^{0.99n}$ (1 mark)

(iii) $256(n+2)(n-1)! + 2^{512n} + n^{1024} + n!$ (1 mark)

(iv) $n^2 + n^{1.9} \log_2 n^{1.8} + 2030$ (1 mark)

(v) $\log_{10} n^{28}$ (1 mark)

(vi) $\log_{100} n^{32} + \log^{32} n$ (1 mark)

(b) Using the definition of Big-Oh, prove $2n^2 = O(n^{2.1})$. (6 marks)
[Hint: $O(f(n)) = \{g(n) \mid \exists c, n_0 \bullet \forall n > n_0 \bullet g(n) \leq c f(n)\}$](c) Using the definition of Ω , prove $2n^2 \neq \Omega(n^{2.1})$. (6 marks)
[Hint: $\Omega(f(n)) = \{g(n) \mid \exists c, n_0 \bullet \forall n > n_0 \bullet g(n) \geq c f(n)\}$](d) Using the definition of Big-Oh, prove $O(n) = O(3n)$. (7 marks)
[Hint: Prove $O(n) \subset O(3n)$, and $O(3n) \subset O(n)$.]

[Total : 25 marks]

UCCC2063 ALGORITHMS ANALYSIS**Q2. [Solving Recurrence]**

- (a) We are given $T(n) = 3T(n/3) + n$. Answer the following questions.
- (i) Prove $\exists k \geq 0 \bullet n = \Theta(n^{\log_3 3} \log^k n)$ (3 marks)
 - (ii) Use master method to solve the recurrence. (3 marks)
- (b) We are given $T(n) = 3T(n/3) + n^2$. Answer the following questions.
- (i) Prove $\exists \epsilon > 0 \bullet n^2 = \Omega(n^{\log_3 3 + \epsilon})$ (3 marks)
 - (ii) Prove $\exists c < 1 \bullet 3f(n/3) \leq cf(n)$, where $f(n) = n^2$. (3 marks)
 - (iii) Use master method to solve the recurrence. (3 marks)
- (c) Consider the recurrence $T(n) = 2T(n/3) + n$.
- (i) Solve $T(n)$ by recursive method (also called expansion). Give your answer in Big-Oh notation. (4 marks)
 - (ii) Solve $T(n)$ by substitution (also called Mathematical Induction). Give your answer in Big-Oh notation. (4 marks)
[Hint: The result must be consistent with (b)(i).]
- (d) Explain briefly why $T(n) = T(n-1) + n$ cannot be solved by master method. (2 marks)
- [Total : 25 marks]

UCCC2063 ALGORITHMS ANALYSIS**Q3. [Binary Tree]**

- (a) We consider rooted binary tree. A leaf node is a node which does not have any children nodes. The height of a binary tree is defined as follows:

The height of an empty tree is defined as 0. For a non-empty tree, the height is defined as the number of nodes in the path from the root to its farthest leaf node.

- (i) Give a recursive definition for the height of a binary tree. (5 marks)
- (ii) Based on (i), give a recursive algorithm, in pseudocode format, for accepting an input binary tree, and returning its height. (4 marks)
- (iii) Analyze the time complexity of your algorithm in (ii). (4 marks)
- (b) A programmer uses the following **struct** for constructing a rooted binary tree.

```
struct node {
    int x;
    node* left;
    node* right;
};
```

Inside a node, **x** is the stored data, **left** and **right** are pointers pointing to the node's left and right children, respectively, if any. If a pointer is not pointing to any node, it is called null pointer (because, the programmer should have assigned it to **NULL**).

Suppose that an n -node binary tree is constructed using the above **struct**. Using induction on n , prove the following statement, for all integers $n \geq 1$.

Statement: An n -node binary tree has $n+1$ null pointers.

(10 marks)

[Hint: The proof should contain base case, and inductive step.]

- (c) A student extends the struct in (b) into the following **struct** for constructing a rooted N -ary tree.

```
struct node {
    int x;
    node* child[N];
};
```

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Q3. (Continued)

As the **struct** in (b), variable **x** is the stored data, the pointers are pointing to children nodes, respectively, if any. If a pointer is not pointing to any node, it is a null pointer.

What is the number of null pointers in an n-node N-ary tree? (2 marks)
[Hint: no need to explain.]

[Total : 25 marks]

UCCC2063 ALGORITHMS ANALYSIS**SECTION B (Choose Any One Question)**

Q4. [Dynamic Programming]

We are given a sequence of positive integers $x_1, x_2, \dots, x_N$, a positive integer N , and the recursive function **sum**(,) below.

```

For  $j < i$ ,
    sum( $i, j$ ) = 0;
For  $j = i$ ,
    sum( $i, j$ ) =  $x_i$ ;
For  $j > i$ ,
    sum( $i, j$ ) = max( $x_i + \text{sum}(i+2, j), x_{i+1} + \text{sum}(i+3, j)$ );
where  $1 \leq i, j \leq N$ .

```

Suppose that the sequence is stored in $X[1..N]$ in order. In pseudocode format, answer the following questions.

- (a) From the recursive function, identify and state all the base cases. (2 marks)
- (b) Give a recursive algorithm, in pseudocode format, for computing **sum**(i, j). (3 marks)
- (c) Let $S[1..N, 1..N]$ be an array which is supposed to store the values of **sum**(i, j), for all i, j between 1 and N , inclusively. Suppose $N=5$, and $x_1=1, x_2=4, x_3=1, x_4=4, x_5=8$. Show the content of the $S[1..N, 1..N]$. (12 marks)
- (d) Give an $O(N^2)$ -time algorithm, in pseudocode format, for computing **sum**(1, N). (8 marks)

[Total : 25 marks]

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Q5. [Graph Algorithms]

Consider an undirected connected graph $G(V,E)$ with n vertices and e edges.

The Breadth First Search (BFS) algorithm is given below.

```

BFS ( $G, s$ )
  for each  $u \in V - \{s\}$  do
     $\text{color}[u] \leftarrow \text{WHITE}$ 
   $\text{color}[s] \leftarrow \text{GRAY}$ 
   $Q \leftarrow \{s\}$ 
  while  $Q \neq \emptyset$  do
     $u \leftarrow \text{head}[Q]$ 
    for each  $v$  in  $\text{Adj}[u]$  do
      if  $\text{color}[v] = \text{WHITE}$  then
         $\text{color}[v] \leftarrow \text{GRAY}$           (*)
        ENQUEUE ( $Q, v$ )
    DEQUEUE ( $Q$ )
     $\text{color}[u] \leftarrow \text{BLACK}$ 

```

The input is (G, s) where G is the graph stored in adjacency list $\text{Adj}[]$, and s is the starting vertex. Q is a queue, and $\text{head}[Q]$ is the first element in Q . **ENQUEUE** and **DEQUEUE** are operations for appending an element to the tail and taking away an element from the front, of the queue, respectively.

- (a) Suppose that every vertex will be enqueued into Q exactly once. Answer the following questions.
- (i) Give an input graph G and a starting vertex s such that Q has exactly n elements in a particular point during the execution. Explain your answer. (4 marks)
 - (ii) Give an input graph G and a starting vertex s such that Q has exactly one element at the beginning of each iteration. Explain your answer. (4 marks)
 - (iii) How many iterations does the while loop have? Give your answer in terms of n and e . Prove your answer. (3 marks)
- (b) Suppose that the colour of each vertex will be turned from white to gray, then from gray to black, and have no further changes. Answer the following questions.
- (i) What is the number of times that line (*) is executed? Prove your answer. (3 marks)

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Q5. (Continued)

- (ii) What is the number of times that the if-condition is executed? Prove your answer. (3 marks)
- (iii) What is the time complexity of **BFS (G, s)** ? Prove your answer. (4 marks)
- (c) A BFS tree is a tree rooted at vertex **s**, and if a white vertex **v** is traversed from a gray vertex **u**, then **u** is the parent of **v** in the tree. Modify the given BFS algorithm to find the BFS tree. (4 marks)
[Hint: Use a 1-dimensional array to store the parents of vertices.]

[Total : 25 marks]
